

Article Summary

Deep Learning Based Brain Tumor Segmentation: A Survey

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1. Purpose and Scope

This survey provides a broad technical review of deep-learning methods for brain-tumor segmentation. The authors examine more than 150 papers and organize the literature around three major themes: **effective segmentation-network design**, **segmentation under imbalanced conditions**, and **learning from multi-modality MRI**. The review also compares methods on public datasets and challenges, especially BraTS, and identifies open research directions.

Brain-tumor segmentation is formulated as voxel- or pixel-level classification of one or more MRI sequences into normal tissue and predefined tumor regions. The paper focuses especially on glioma, including low-grade glioma (LGG) and high-grade glioma (HGG), and on clinically meaningful subregions such as enhancing tumor, peritumoral edema, and necrotic/non-enhancing tumor core.

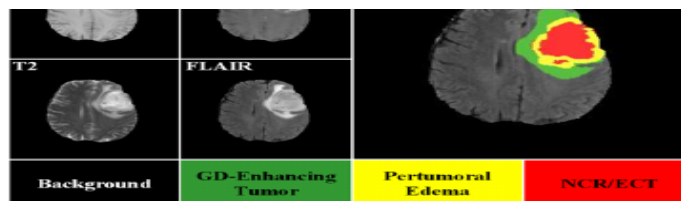


Fig. 2 Exemplar input dataset with different MRI modalities and corresponding ground truth segmentation map. Each frame represents a unique MRI modality. The last frame on the right is the ground truth with corresponding manual segmentation annotation. Different colors represent different tumor sub-regions, i.e., gadolinium (GD) enhancing tumor (green), peritumoral edema (yellow) and necrotic and non-enhancing tumor core (NCR/ECT) (red)

MRI sequences), the system aims to automatically segment the tumor area from the normal tissues by classifying each voxel or pixel of the input data into a pre-set tumor region categories. Finally, the system returns the segmentation map of

Article Fig. 2 (p. 2). Example multimodal MRI input (T1, T1ce, T2 and FLAIR) and the manual ground-truth segmentation of glioma subregions.

2. Why the Problem Is Difficult

The survey identifies five central difficulties: **location uncertainty**, because gliomas can arise throughout the brain; **morphological uncertainty**, because tumor size, shape and edema vary greatly; **low contrast** and blurred tissue boundaries in MRI; **annotation bias** caused by differences among expert annotators; and **data imbalance**, because tumor subregions occupy very different numbers of voxels.

3. Data Imbalance and Tumor Heterogeneity

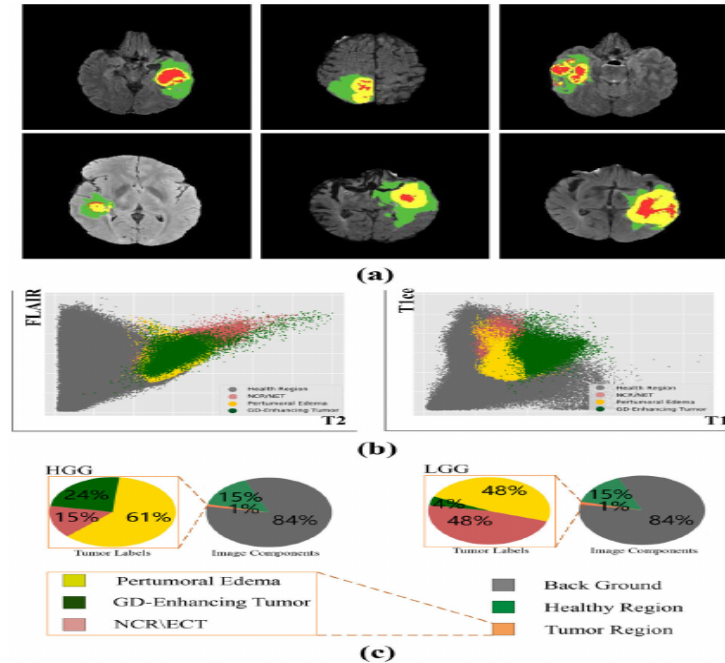


Fig. 3 Challenges in segmentation of brain glioma tumors. **a** Shows glioma tumor exemplars with various sizes and locations inside the brain. **b, c** Show the statistical information of the training set in the multimodal brain tumor segmentation challenge 2017 (BraTS2017). The left hand side of **b** shows the FLAIR and T2 intensity projection, and the right hand side shows the T1ce and T1 intensity projection. **c** is the pie chart of the training data with labels, where the top figure shows

Article Fig. 3 (p. 4). Tumors vary markedly in size and location. The accompanying BraTS2017 distributions illustrate intensity overlap and strong class/region imbalance.

The imbalance problem occurs at several levels. Background and healthy tissue can dominate the image; within the tumor, smaller regions such as necrotic/non-enhancing core may be underrepresented; and the numbers of HGG and LGG patient cases can differ substantially. Consequently, a network trained naively may favor dominant classes and learn less effectively from small but clinically important structures.

The survey groups imbalance remedies into three broad strategies. **Cascade and ensemble networks** combine multiple predictors or progressively segment whole tumor, tumor core and enhancing tumor. **Multi-task learning** introduces auxiliary objectives so representations capture additional useful structure. **Customized loss functions**, including Dice-related and weighted losses, increase the influence of difficult or minority samples. The authors emphasize that ensembles can improve performance but increase computational demand.

4. Evolution of Brain-Tumor Segmentation

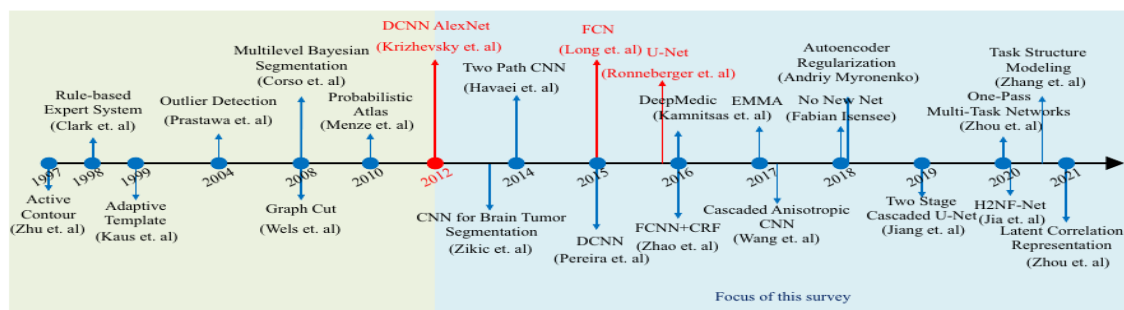


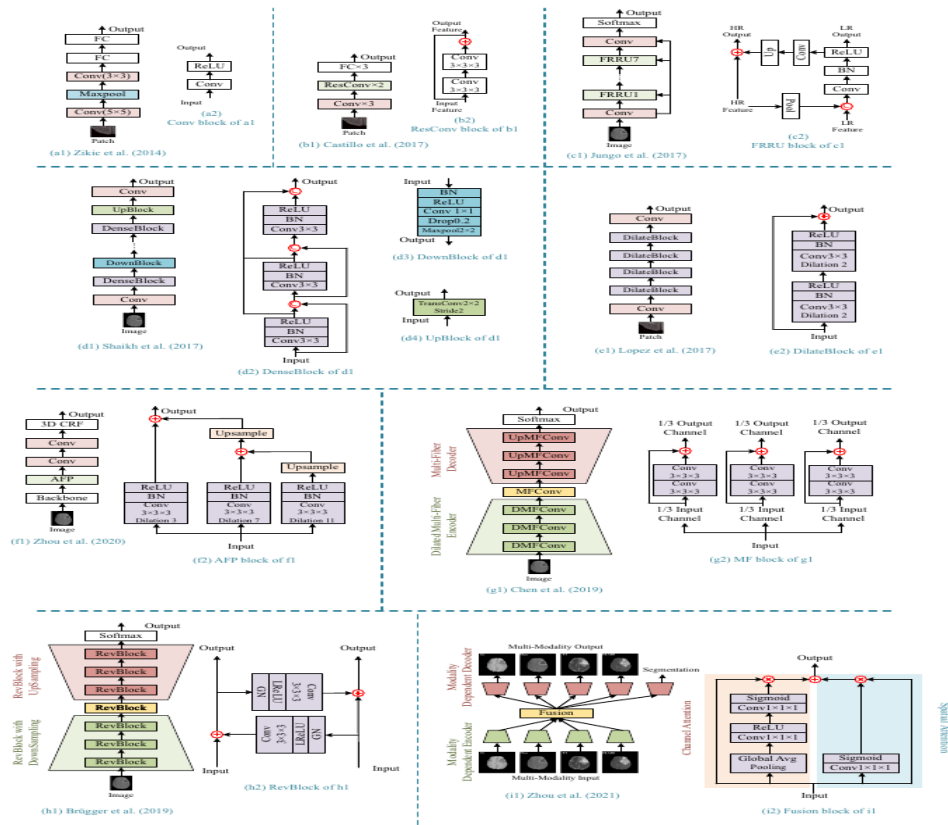
Fig. 4 The evolution of brain tumor segmentation with selective milestones over the past decade. Best viewed in colors

Article Fig. 4 (p. 5). Selected milestones from rule-based and probabilistic methods through AlexNet-era DCNNs, FCNs, U-Net, DeepMedic, cascaded networks, autoencoder regularization and later multi-task/fusion systems.

5. Designing Effective Deep Networks

The paper describes two complementary design goals: improve **accuracy** by learning high-level semantics while preserving precise localization, and improve **efficiency** by reducing parameters, memory use and inference time. Common mechanisms include residual and dense connections, dilated convolution, multi-scale processing, attention and feature fusion.

Small 3x3 kernels became common because stacked small kernels can achieve large receptive fields with fewer parameters than a single large kernel. Residual connections help stabilize deeper networks and mitigate vanishing/exploding gradients. Dilated convolutions enlarge receptive fields without proportionally increasing parameters or sacrificing spatial resolution, while multi-scale dilation helps capture both large edema regions and subtle structures.



structural comparison between representative methods based on effective network modules and architectures. From top-left right: (a1) CNN in [160], (b1) CNN with (b2) residual condude [19], (c1) CNN with (c2) full resolution residual unit

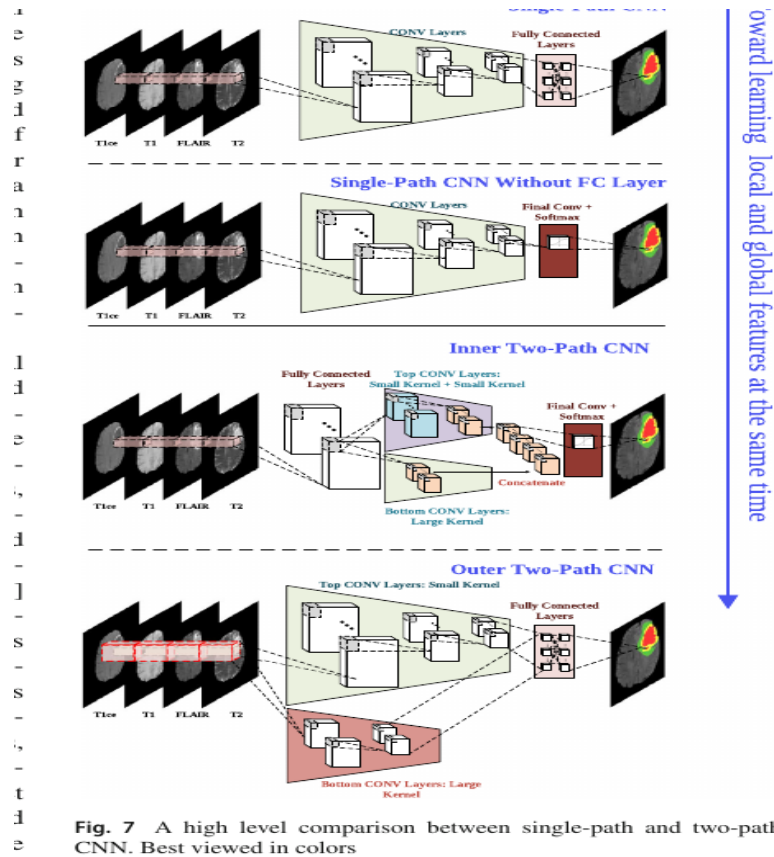
with (e2) residual dilation block [91], (f1) CNN with (f2) atrc volution feature pyramid module [155], (g1) FCN with (g2) mu unit [22], (h1) FCN with (h2) reversible block [14] and (i1) FC (i2) modality fusion module [153]. Best viewed in colors

Article Fig. 6 (p. 8). Structural comparison of representative modules and architectures, including residual, dense, dilated, multi-fiber, reversible and modality-fusion designs.

Efficiency is particularly important for 3D MRI. The survey discusses lightweight U-Net variants, reversible blocks that reduce activation-memory requirements, depthwise-separable convolutions, and architectures such as DMFNet that aim to retain spatial information while lowering computational cost.

6. From Patch-Based CNNs to Encoder-Decoder Networks

Early single-path CNNs process local patches efficiently, whereas **multi-path CNNs** combine pathways at different spatial scales. Large-scale paths tend to capture global information such as tumor location, size and shape, while smaller-scale paths preserve local texture and boundary details.



Multi-path architecture

Article Fig. 7 (p. 9). High-level comparison of single-path and two-path CNN designs for jointly learning local and global features.

Patch-based methods have important limitations: performance depends on patch size and quality; large patches increase computational cost; and fully connected mappings do not provide an ideal end-to-end image-to-mask formulation. The field therefore shifted toward **fully convolutional encoder-decoder architectures**, particularly FCN and U-Net families. Encoders extract increasingly abstract features, while decoders restore spatial resolution and produce dense segmentation maps. U-Net skip connections directly transfer encoder features to matching decoder stages, helping recover fine spatial detail.

The survey concludes that custom modules can improve both speed and accuracy and that specialized architectures can learn features at multiple scales. At the same time, architecture design remains strongly dependent on human expertise, motivating future neural architecture search and stronger incorporation of tumor-specific domain knowledge.

7. Exploiting Multi-Modality MRI

MRI modalities emphasize different tissue properties, so combining T1, T1ce, T2 and FLAIR can improve tumor delineation. The survey organizes multi-modality learning around **ranking modalities**, **pairing modalities**, **feature fusion**, **generating missing modalities**, and **fusion when modalities are missing**.

Modality pairing can identify useful combinations but exhaustive pairing is computationally expensive. Feature fusion generally improves representation and generalization. Simple addition or concatenation is inexpensive but may not model richer cross-modality relationships; learned fusion and attention can strengthen feature expression at the cost of additional parameters. Missing modalities are especially important clinically, because complete high-quality modality sets are not always available. Generative approaches attempt to synthesize missing sequences, although their output quality depends heavily on the available source modalities.

8. Evaluation and BraTS

The survey relies heavily on the **Multimodal Brain Tumor Segmentation Challenge (BraTS)** as a common benchmark. It compares methods using tumor-region measures such as whole tumor (WT), tumor core (TC) and enhancing tumor (ET). **Dice overlap** is a principal accuracy measure, while Hausdorff distance is used to assess boundary disagreement. The review stresses that direct comparison must consider dataset split and evaluation protocol, not just headline scores.

9. Future Research Directions

Less supervision: Fully supervised segmentation depends on expensive, time-consuming expert masks that can contain annotation bias. The authors expect greater use of self-supervised, weakly supervised and semi-supervised learning so useful representations can be learned with fewer labels.

Neural architecture search and domain knowledge: Because module and architecture design still relies heavily on human experience, the survey anticipates combining neural architecture search with brain-tumor domain knowledge, including tumor grade and morphology, to discover more effective architectures.

Patient privacy: As models increasingly exploit imaging and patient information, particularly for downstream tasks such as survival prediction, the authors identify privacy-preserving learning frameworks as an important direction.

10. Overall Takeaway

The paper presents deep-learning brain-tumor segmentation not as a single architecture problem but as the interaction of **network design**, **class imbalance**, **multimodal information** and **clinical data constraints**. Progress from handcrafted pipelines to end-to-end FCN/U-Net systems has substantially strengthened feature learning, but robust segmentation still depends on handling heterogeneous tumor morphology, minority subregions, missing modalities, computational limits and imperfect annotations.

The most useful organizing principle from the survey is its three-part taxonomy: build stronger and more efficient architectures; explicitly address imbalance rather than treating every voxel equally; and exploit complementary MRI modalities while remaining robust when some modalities are unavailable.

Source: Liu Z, Tong L, Chen L, et al. *Deep Learning Based Brain Tumor Segmentation: A Survey*. Complex & Intelligent Systems. 2023;9:1001-1026.